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ORIGINAL ARTICLE

The evaluation of convectively coupled waves in limited-length datasets

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Abstract

Organised tropical convection is strongly influenced by intraseasonal convectively coupled waves (CCWs), particularly Kelvin waves and the Madden-Julian Oscillation (MJO). Kilometre-scale models simulating deep convection explicitly offer new possibilities for investigating the multiscale dynamics of these waves. Due to the high computational cost of running kilometre-scale models, the output is often limited in duration. To evaluate the uncertainties in detecting CCWs using FFT-based and wavelet-based methods from datasets shorter than one year, we employ a measurement known as the Signal Ratio (SR), defined within the zonal wavenumber-frequency spectrum. The wavelet-based detection method slightly outperforms the FFT-based method in identifying CCWs across dataset lengths. Detectability is affected by the variable used: OLR provides superior detection for Kelvin waves, while precipitation is more useful for the MJO. Applying the wavelet-based method, Kelvin waves can be robustly detected with at least 90 days. The MJO can be robustly detected with at least 30 days of precipitation data. A SR-based criterion is applied to a case study of six tropical kilometre-scale simulations by the Unified Model (UM). Across the simulations, skill in representing Kelvin waves and the MJO appears linked, suggesting that both can be affected by similar aspects of model design. Our results encourage the use of wavelet-based detection, especially with relatively short datasets, and emphasize the importance of selecting suitable variables for detection. They also provide a benchmark for CCWs in numerical model runs of limited duration.

Keywords Convectively coupled waves · MJO · Wavelet · Convection-permitting models (CPMs)

1 Introduction

Organised deep convection in the tropics is often modulated by equatorial atmospheric wave-like motions, collectively referred to as convectively coupled waves (CCWs). These include Convectively Coupled Equatorial Waves (CCEWs) predicted by linear theory, such as Kelvin and equatorial Rossby waves (Matsuno 1966; Takayabu 1994; Kiladis et al. 2009), as well as non-linear planetary-scale modes like Madden–Julian Oscillation (MJO) (Madden and Julian 1971, 1972, 1994) and synoptic-scale distur-bances like Tropical-Depression-Type Waves (Takayabu and Nitta 1993). The MJO and Kelvin waves are two of the phenomena having the most influence on tropical intraseasonal precipitation variations (Worku et al. 2020; Schreck 2015) as well as extreme rainfall events in regions like tropical Africa (Kebacho et al. 2025; Berhane et al. 2015), the Maritime Continent (Latos et al. 2021; Doyle et al. 2023; Senior et al. 2025), and Australia (Whelan and Frederiksen 2015; Marshall et al. 2023; Muhammad et al. 2024).

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Their coupling with deep convection makes the MJO and Kelvin waves ideal testbeds for studying moisture-convection-circulation interactions. Therefore, this study focuses on these two waves. Over six decades of research, observational and theoretical studies have consistently demonstrated that moisture availability is a critical factor in controlling convective development. At the same time, latent heating from convection acts as a key driver of atmospheric wave dynamics (Morita et al. 2006; Xu et al. 2015; Mayta et al. 2021). For instance, Johnson and Ciesielski (2013), using data from the DYNAMO campaign, demonstrated that deep convection in the MJO is preceded by a gradual moistening of the lower troposphere, suggest-ing a recharge–discharge mechanism for convection initiation. Similarly, Kikuchi and Takayabu (2004) documented the stepwise evolution of cloud populations during MJO passage in the TOGA COARE field campaign, from shallow cumulus to deep con-vective systems, tied to increasing tropospheric humidity. In the case of convectively coupled Kelvin waves (CCKWs), Straub and Kiladis (2003) confirmed that the verti-cal structure and phase tilt of wave-related temperature and wind fields are consistent with heating from organised deep convection. These findings collectively reveal that the tropical atmosphere can self-organise across scales—from mesoscale convective systems to planetary-scale waves—through multiscale feedbacks between convection, circulation, and moisture. Recent studies, such as Mayta et al. (2021), further highlight that the MJO behaves as a moisture mode over warm pool regions, where convection is tightly coupled to column-integrated humidity, but transitions to a more wave-propagation dynamical characteristics over the central Pacific. Insights from such research have led to significant advances in the parameterisations of general circulation models (GCMs) (Hung et al. 2013; Wang et al. 2019), improving the simulation of precipitation variability, monsoon onset, and even tropical cyclone modulation in subseasonal forecasts (Jung et al. 2010; Camargo et al. 2020).

By allowing for the direct simulation of deep convection, convection-permitting models (CPMs) hold the promise of simulating CCWs more realistically via explicit representation of the interactions between large-scale dynamics and convection (Kang et al. 2016, 2019; Zheng et al. 2024). Beginning from the 1990s, field campaigns and research on explicit cloud physics have supported the development of CPMs (Gottschalck et al. 2013; Chen et al. 2015). The more realistic diabatic heating pro-files from CPMs may account for the improved simulation of the vertical structure of water vapour, and wind anomalies caused by the waves (Rios-Berrios et al. 2023). The stronger heating below the melting level during the active convection phase in CPMs can, for example, enhance the interaction between shallow convection and low-level moisture convergence, yielding better MJO simulations (Liu et al. 2022; Rios-Berrios et al. 2023).

However, the ability to detect CCWs in the limited output of presently available kilometre-scale models (K-models) is unclear. Multi-year kilometre-scale simulations are limited because of the high resolution and large domains needed, and most of the available simulations are relatively short. Although the output length has increased over the last decade from a few months to around one year (Sch¨ar et al. 2020; Judt and Rios-Berrios 2021; Takasuka et al. 2024; Rackow et al. 2025), such data lengths still limit the application of detection methods for CCWs. Previous work holds that two general CCW detection methods, based either on a narrow filter window (based on FFT or Wavelet transform) or a broad one (via spatial projection), can iden-tify CCWs with multi-year data, such as over 40-year observed outgoing longwave radiation (OLR) records (Wheeler and Kiladis 1999; Knippertz et al. 2022). The uncer-tainties in using them to detect CCWs in datasets with short periods, however, remain unclear.

The Combined FFT-Wavelet Transform (CFWT) is a method based on the wavelet transform that detects the evolution of waves, first proposed by Kikuchi (2014). Compared with other methods based on the Fourier transform that have a narrow filter window, such as the Wheeler-Kiladis method (WK1999) (Wheeler and Kiladis 1999), CFWT replaces the Fourier transform along the time dimension with the wavelet transform, which means it can capture local wave information and track temporal changes in waves (Domingues et al. 2005; Kikuchi and Wang 2010). This change aban-dons the assumption that CCWs are stationary signals, which can, for example, allow for the Doppler shifting or other speed or amplitude variations in observed CCWs (Yang et al. 2003, 2007). Knippertz et al. (2022) compare several objective CCW detection methodologies—emphasizing spatial structure, propagation characteristics, and robustness to short records—yet they do not conclusively assess the merits of wavelet-based over FFT-based detection methods.

This study evaluates uncertainties in applying wavelet-based and FFT-based detec-tion methods, and clarifies the record lengths required to detect CCWs robustly. Specifically, we address (1) whether a wavelet-based method, CFWT, can outperform the FFT-based method (in this study, we applied WK1999), WK-FFT, or bring other advantages in detecting CCWs in a short dataset; (2) how long a dataset must be to detect CCWs robustly or measure their fidelity to observations; (3) what's the defini-tion of a robust or good detection of CCWs? Finally, we apply our methods to a short simulation case—the kilometre-scale Unified Model (kilometre-scale UM).

2 Methodology

2.1 Data

2.1.1 Observations

Observational precipitation data from Integrated Multi-satellitE Retrievals for GPM (IMERG) and outgoing longwave radiation (OLR) data from NOAA (Hai-Tien and Program 2011) are used to test the CCW detection methods and to quantify the criteria of "robust detections". We use the 20-year period from 1st January 2001 to 31st December 2021. Gridded IMERG data at one-degree resolution are obtained from the Frequent Rainfall Observations on GridS (FROGS) database (Roca et al. 2019).

2.1.2 Kilometre-scale UM simulations

To illustrate the methods, we apply them to simulations of the Kilometre-scale UK project from the Met Office. By applying the UM seamless modeling framework, this UK project runs global (GLOB) and cyclic tropical channel (CTC) simulations with the same code. The simulations are essentially downscaling runs, with a coarser global model downscaled in the tropical belt to a finer grid. Simulations were run during the DYAMOND-winter period (20th January to 28th February 2020).

The atmosphere-only global simulations are run without data assimilation; they are started from Met Office global analyses. Sea surface temperatures are prescribed using Operational Sea Surface Temperature and Sea Ice Analysis (OSTIA) and updated daily throughout the simulations. The global simulations employ an n1280 latitude–longitude grid with a resolution of approximately 15 km × 10 km in the tropics (GLOB10), and utilize 70 vertical levels with a model top at 80 km. To test the role of lateral boundary conditions, GLOB10 is run with convection parameterization, called Global-atmosphere-land (GLOB10 GAL9) (Walters et al. 2019) and explicit convection, that is, no deep convective parameterization, named Regional-atmosphere-land (GLOB10 RAL3) (Bush et al. 2025). For each of these two global forcings, three configurations for the CTC runs were done: a 4.4 km explicit convection run (CTC4 RAL3), a 5.5 km (n2560) explicit convection run (CTC5 RAL3), and a 5.5 km convection parameterisation run (CTC5 GAL9). In this study, we examine each of these six downscaled simulations, as well as the two global driving simulations.

2.1.3 Data sampling

The sample data used for estimating confidence intervals (CI) of statistical measures and red noise comes from a random sampling of observational datasets. To test for each L, we obtain an ensemble of time segments of length L from the 20-year observational records N starting from randomly selected time points tr. The data sample $L_{tr} \in N$ is defined as the data segment from tr to $tr + L - 1$ in N after calculating anomalies.

2.2 The combined FFT-wavelet transform

By applying wavelet analysis, the CFWT provides information on the changes of each wave over time and can evaluate the signal for every frequency. To stabilise the wavelet transform near the edges, the time series L was symmetrically padded to the next power-of-two length ($2^{[log_2 L]}$). The explanation and parameter settings for this method come from Kikuchi (2014) (Eqs 1 to 5).

The forward calculation of the CFWT from the physical field to the scale–wavenumber domain (domain after forward CFWT) is given by

$$(mathcalTg)(k,s,\tau) = \frac{1}{2\pi}\int_0^{2\pi} dx \int_{-\infty}^{\infty} g(x,t)\psi^*_{s,\tau}(t)e^{ikx}dt \qquad (1)$$

To obtain $\psi_{s,\tau}(t)$, let ψ denote the mother wavelet, here chosen to be the complex Mor-let wavelet $\psi(t) = e^{i\omega_0 t}e^{-t^2/2}$ (Meyers et al. 1993). Its scaled and shifted form is given by $\psi_{s,\tau}(t) = |s|^{-1/2}\psi\left(\frac{t-\tau}{s}\right)$, where the scale parameter $s \in \mathbb{R}, s \neq 0$ controls the temporal dilation, and the translation parameter $\tau \in \mathbb{R}$ determines the position in time. The asterisk indicates the complex conjugate, and k denotes the zonal wavenumber. The CFWT is also reversible:

$$g(x,t) = \frac{1}{C_\delta}\sum_{k=-\infty}^{\infty}\int_{-\infty}^{\infty}(mathcalTg)(k,s,t)e^{-ikx}\frac{ds}{|s|^{3/2}} \qquad (2)$$

where $C_\delta = \sqrt{2\pi}\int_{-\infty}^{\infty}\frac{\left|\widehat{\psi}(\omega)\right|}{\omega}d\omega$. From this inverse equation, a spatial–temporal filter can be defined as

$$g'(x,t) = \frac{1}{C_\delta}\sum\nolimits_D\int(mathcalTg)(k,s,t)e^{-ikx}\frac{ds}{|s|^{3/2}} \qquad (3)$$

where $D = \{(k,s) \in \mathbb{R}^2\}$ is a selected region in the scale–wavenumber domain, and $g\prime(x,t)$ is the filtered signal reconstructed by including only the selected part of the spectrum.

Wavelet analysis on each longitude in the domain, after applying the Fourier trans-form to each time step, reveals a relationship between the scale s and the Fourier frequency f ($f = \omega/2\pi$):

$$s = \frac{\omega_0 + \sqrt{\omega_0^2 + 2}}{4\pi f} \tag{4}$$

This relationship serves as a reference for the transformation from the scale-wavenumber domain to the Fourier domain (frequency-wavenumber domain), which forms the foundation for comparing the detection method of CCWs in the frequency-wavenumber spectra.

The CFWT maintains energy conservation in both forward and backwards processes through standardisation (Parseval's relation),

$$\begin{aligned} &\frac{1}{C_\psi} \sum_{k=-\infty}^{k=\infty} \int_{-\infty}^{\infty} \frac{ds}{s^2} \int_{-\infty}^{\infty} |(mathcalTg)(k,s,\tau)|^2 d\tau \\ &= \int_0^{2\pi} dx \int_{-\infty}^{\infty} |g(x,t)|^2 dt = 2\pi \sum_{k=-\infty}^{k=\infty} \int_{-\infty}^{\infty} |G(k,\omega)|^2 d\omega \end{aligned} \tag{5}$$

where $C_\psi = \int_{-\infty}^{\infty} \frac{\left|\widehat{\psi}(\omega)\right|^2}{\omega} d\omega$.

We have built a Python package that incorporates the above transformations and filtered signal reconstruction (see end of paper for details on where to obtain this).

2.3 Hypothesis test

We present an approach to quantify the intensity of CCWs and a statistical method to quantify the detection of CCWs. This enables us both to compare the detection methods of CCWs and to set a quantifying criterion to determine whether simulations yield reasonable signals of CCWs when applied to observational data.

The Signal Ratio (SR) is defined as the ratio of CCW energy to the total energy in the Fourier domain F, where the numerator is the spectral energy contained within a selected CCW band U (zonal wavenumber k and frequency f (in cycles per unit time, e.g., days in this research) ranges following a theoretical dispersion relation). The denominator is the energy integrated over all available spectral bins resolved by the CFWT (domain T), covering all wavenumbers k and positive frequencies f up to the Nyquist limit. Coefficients outside the cone of influence (COI), including all padded-only regions, were discarded when computing SR. The SR in a time period L is defined as:

$$\mathrm{SR}_L = \frac{E_U}{E_T} \tag{6}$$

where

$$E_U = \iint_{(k,f)\in U} |G(k,f)|^2 dk df \tag{7}$$

is the integrated spectral energy of the target wave mode (e.g., Kelvin, MJO) in a period L, obtained by applying a window for target frequency $(f \in U)$ and target zonal wavenumber $(k \in U)$ in F, and

$$E_T = \iint_{(k,f)\in T} |G(k,f)|^2 dk df \tag{8}$$

is the spectral energy in T.

We develop a statistical method to determine robust detection by applying the sample SR as a test statistic, finding a "percentile range of SR", which determines (for a specific data sample) whether a measured SR might have occurred by chance and (in general) how accurately we can quantify CCW activity. To do this we compute the range of likely sample SR values under, first, a null hypothesis (red noise); and second, a "truth" hypothesis (observed data). For each data sample (Sect. 2.1.3), SR is computed from the spectra calculated using both WK-FFT and CFWT ($G_{\mathrm{obs}}(k, f)$) and the red noise spectra calculated from ($G_{noise}(k, f)$) by the first-order auto-regressive model (AR(1)). Although CFWT and WK-FFT have different time–frequency resolution trade-offs, the SR values remain directly comparable because both methods apply the same fixed $k - f$ filtering windows to make sure wave and background variance are integrated over equivalent spectral regions.

The calculation of the $G_{noise}(k, f)$ based on AR(1) comes from Vaughan et al. (2011). It is a spectra-fitting AR(1) method, which takes the already computed power spectra $G_{\mathrm{obs}}(k, f)$ and fits $G_{\mathrm{noise}}(k, f)$ directly to each zonal wavenumber's frequency spectra $G_{\mathrm{obs}}(f)$:

$$G_{\mathrm{noise}}(f; N, \alpha) = N \cdot \frac{1 - \alpha^2}{1 - 2\alpha \cos\left(\frac{\pi f}{f_N}\right) + \alpha^2 + \varepsilon} \tag{9}$$

where f is frequency, $f_N = \frac{1}{2\Delta t}$ is the Nyquist frequency, N is the normalisation factor (mean power), α is the AR(1) lag-1 autocorrelation coefficient, $\alpha \in (-1,1)$ and ε is 0.001 a small constant added to prevent division by zero.

Following the idea from Stella et al. (1994) and Vaughan et al. (2011), the AR(1) parameters (α, N) from Eq. (9) are determined by maximum likelihood estimation for each zonal wavenumber by minimising the negative log-likelihood under a Gamma distribution for power values via

$$(N^*, \alpha^*) = \arg \min_{N>0, \alpha \in [\alpha_{\min}, \alpha_{\max}]} (-2 \log mathcalL(N, \alpha)) \tag{10}$$

$$\begin{aligned} &\log mathcalL(N, \alpha) \\ &= \sum_f \left[(\nu_f - 1) \log G_{\mathrm{obs}}(f) - \nu_f \log\left(\frac{G_{\mathrm{noise}}(f)}{\nu_f}\right) - \log \Gamma(\nu_f) - \frac{G_{\mathrm{obs}}(f)}{G_{\mathrm{noise}}(f)/\nu_f} \right] \end{aligned} \tag{11}$$

where $v_f = 2\sqrt{1+\left(\frac{1.03 f N_\alpha(f) dt}{\Gamma}\right)^2}$ is the effective degrees of freedom at each frequency f, which accounts for both the sample size after removing the cone of influence and the decorrelation properties of the wavelet. $\Gamma(\cdot)$ is the Gamma function $\Gamma(v) = \int_0^\infty t^{v-1} e^{-t} dt$.

The background red-noise spectrum $G_{\text{noise}}(k, f)$ for each zonal wavenumber k is.

(Kikuchi 2014)

$$G_{\text{noise}}(k, f) = G_{\text{noise}}(f; N^*(k), \alpha^*(k)) \tag{12}$$

Now, for both $G_{\text{obs}}(k, f)$ and $G_{\text{noise}}(k, f)$, we applied the same statistical method to determine whether the SR is reasonable from certain data with length L. Let SR_i denote the SR of the i-th sample ($i = 1, 2, 3, \ldots, 1000$). Then $[SR_{\text{obs}}]_{5\%-95\%}$ and.

$[SR_{\text{noise}}]_{5\%-95\%}$ are defined as the empirical 5th and 95th percentiles of the respective SR distributions, which are called "percentile range of SR" for observational data and for red noise, respectively.

We define detection as robust when the 50th percentile of $[SR_{\text{obs}}]_{50\%}$ is above the 95th percentile of $[SR_{\text{noise}}]_{95\%}$ (Wheeler and Kiladis 1999). Otherwise, it means the data length is too short to distinguish CCW. This means we are insisting on at least a 50% chance of obtaining a detection at 95% confidence. If a model falls within the red noise range, then we can't identify any CCW (and the top of this range could be defined as a detection percentile range). If it falls within the "truth" range, then it is consistent with observation (but if the range is wide, this is not a very strong statement).

When applying CFWT, significance levels for MJO detection in short records should be interpreted with caution since the significance testing against red noise still assumes approximate stationarity, although CFWT is well-suited for analysing nonstationary signals. For Kelvin waves, a 20–40 day record contains several cycles, so the AR(1) assumption is less restrictive. For the MJO, however, records shorter than 90 days may include only about one cycle, making the stationarity assumption questionable and the red-noise confidence intervals less robust (Maraun et al. 2007).

3 Results

3.1 CCW Detection in short-term datasets

Figure 1 presents the SR as a function of window length, for CFWT-based and WK-FFT-based methods, in detecting Kelvin waves and the MJO, applied both to precipitation (IMERG; blue) and OLR (NOAA; red) data. The analysis spans window lengths from 20 to 910 days, as detection using data lengths below 20 days was judged a priori to be unreliable, because spectral estimation from very short records suffers from large sampling uncertainty and strong background noise compared to typical wave signal amplitude. For equatorial Kelvin waves, whose spectral amplitude in the 2–20-day band is often modest compared to background weather fluctuations, such short records are therefore insufficient for robust detection (Das et al. 2021).

OLR exhibits systematically higher SRs than precipitation, which is consistent with the stronger spatial coherence captured in infrared (high cloud) measurements compared to noisy precipitation fields. In addition, the SR of Kelvin waves is higher than the corresponding SR of the MJO,

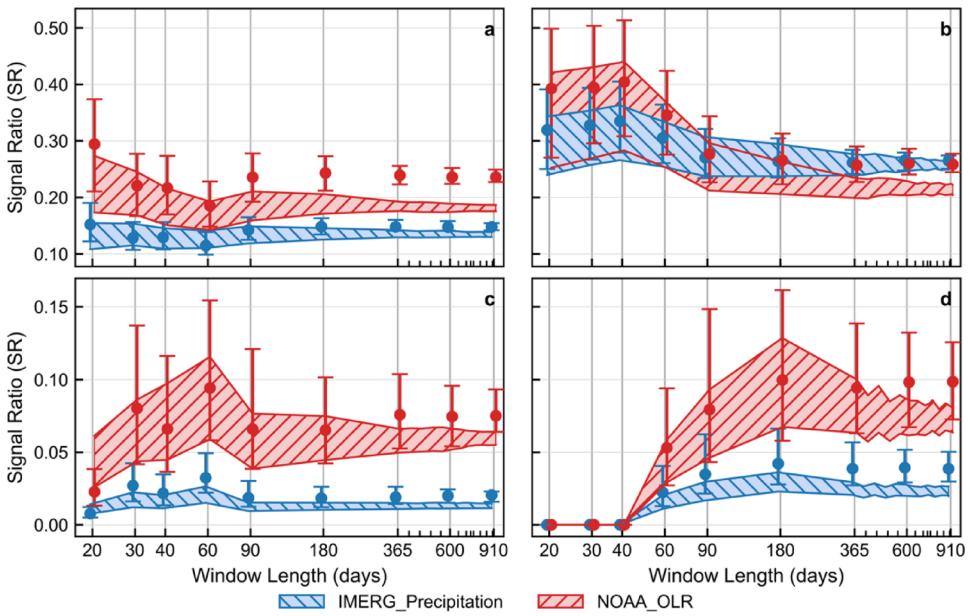


Fig. 1 Sample-measured SR using CFWT **a** & **c** and WK-FFT **b** & **d** methods for equatorial waves. The top panels **a**,**b** and bottom panels **c**,**d** represent results for Kelvin waves and the MJO, respectively. In each panel, the red series represents results based on OLR, and the blue series represents results based on precipitation. The shaded areas represent the 5-95-percentile range of SR from red noise, while error bars show the same percentile range from observations, with the median shown by a dot. The analysis domain spans the tropical belt from 20° S to 20° N across all longitudes. The separation between shading and error bar indicates detectability (see text)

which comes from a larger filter window (zonal wavenumber: 1 to 14, period: 2.5 to 20) of the Kelvin waves compared to that of the MJO (zonal wavenumber: 1 to 5, period: 30 to 96). Although higher SR values suggest stronger wave signals relative to the entire spectrum, signal detectability must consider whether the wave signal is significant in relation to the associated red noise. For example, Kelvin wave detection performs worse than MJO detection in the same 40-day datasets due to the lower signal-to-noise ratio of Kelvin waves. In the 2–20-day frequency band relevant for Kelvin waves, OLR variability is dominated by upper-tropospheric cloud effects and other background weather fluctuations that contribute substantial red-noise power. Consequently, even when Kelvin waves are present in the dataset, the OLR-based SR in short segments (e.g., 40 days) can remain statistically indistinguishable from red noise due to limited sampling and strong background variance. In contrast, the MJO's longer period and larger amplitude allow its SR to emerge above the noise level within the same time window.

The CFWT meets our detection condition for Kelvin waves from OLR with record lengths of 90 days or more, while detectability is only marginal with 20–60 days of data. For precipitation, detectability is absent for records shorter than 90 days, marginal at 90 days, and becomes clear only at 180 days (Fig. 1a). We note that the fluctuations in the relative position of individual median points with respect to the red-noise trend reflect the uncertainty arising from finite observational records. In other words, the true median SR cannot be determined exactly from limited samples. Thus, individual points may lie slightly above or below the detection percentile range even when the overall tendency is clear. The WK-FFT approach, by contrast, requires at least 180 days of OLR data or one year of precipitation data to detect Kelvin waves robustly (Fig. 1b). In contrast to Kelvin waves, the MJO can be robustly detected from precipitation data of at least 30 days via CFWT, while it requires at least 365 days of OLR (Fig. 1c). Due to the more limited frequency resolution, WK-FFT requires at least 60 days of precipitation data to detect the MJO robustly, and it cannot offer a robust detection from OLR even with one year of data (Fig. 1d). Therefore, CFWT can provide robust detections in short datasets (data length $\leq$ one year) compared with WK-FFT, and is the only method that can robustly detect both Kelvin waves and the MJO from either OLR or precipitation data.

The analysis above demonstrates that the data type can influence the minimum length required for robust detection of CCWs. OLR data are better for detecting Kelvin waves, while precipitation data can provide a better detection of the MJO. The clearer distinction between Kelvin waves and the MJO in observational spectra is primarily due to their detectability in different datasets. Cho et al. (2004) reveals that the Kelvin wave has a higher signal-to-noise ratio in OLR since the high noise from the highly localized convective precipitation in the short-period band reduces the statistical significance of the Kelvin wave. For the MJO, the precipitation field often exhibits large-scale and significant rainfall anomalies, so the MJO signal is very distinct in the precipitation-based spectra (Nakazawa 1988; Kim and Zhang 2021). The choice of OLR to define and track the MJO in early studies may have been motivated by OLR's long, continuous satellite record and global coverage from the 1970s onward, as well as the prominent high-cloud signals associated with deep convection. These characteristics enabled OLR to successfully reveal the structure of tropical waves in early research.

The median SR is not always period-invariant as might be expected. For Kelvin waves, the median SR decreases as the data length increases from 20 to 60 days. With CFWT, OLR-based SR peaks at ~0.3 for 20–40-day records and declines to ~0.25 at longer windows, while precipitation-based SRs remain below 0.2 across all lengths (Fig. 1a). The WK-FFT method shows a similar peak (OLR ~0.4, Precip ~0.33) between 20 and 40 days, with differences between data types diminishing beyond 90 days (Fig. 1b). These trends reflect the narrowing of the available spectral window or loss of low frequencies in shorter records, which differently affects the numerator and denominator in Eq. (6).

For long records, however, CFWT yields robust detection from either OLR or precipitation for both Kelvin waves and the MJO once the record length exceeds the stricter of the two data-type thresholds (~180 days for Kelvin and ~365 days for the MJO). By contrast, WK-FFT remains data-type dependent to longer windows: data-type independence emerges at ~365 days for Kelvin and ~450 days for the MJO (OLR). Beyond these record lengths, results are robust regardless of whether OLR or precipitation is used, so the dataset choice can then be based on availability or other practical considerations, while for shorter records, the choice of dataset remains critical.

The percentile range of SR narrows progressively with increasing record length,

consistent with the expected $1/\sqrt{N-1}$ dependence on effective sample size (Fig. 1).

This reflects reduced sampling uncertainty at longer record lengths, as more independent realizations of wave cycles become available. To provide a stable benchmark for evaluating simulations, we define "standard SR" $[SR_{\text{obs}}]_{50\%}$ as the average of the median SR values from the longest available observational records. A narrower percentile range indicates high confidence in this benchmark, whereas a wider percentile range reflects greater variability and thus larger uncertainty in CCW detection. Therefore, with a larger percentile range, WK-FFT indicates greater

uncertainty in detecting CCWs than the CFWT. WK-FFT spectral estimates are based on global frequency–wavenumber averages, and thus their statistical variability increases with more stringent (higher) percentile thresholds, leading to greater uncertainty in detection. In contrast, CFWT's time–frequency localization yields more stable signal estimates and lower detection uncertainty. For the CFWT method, the SR standard of Kelvin wave is 0.236 (percentile range: 0.227 to 0.250) from the OLR-based spectra, and 0.148 (percentile range: 0.142 to 0.155) from the precipitation-based spectra. The OLR-based SR standard of the MJO is not as clear as that based on precipitation data. For the MJO, the standard SR is 0.076 (0.064 to 0.093) from OLR data and 0.020 (0.014 to 0.026) from precipitation. WK-FFT generally yields higher SR reference values than CFWT across most window lengths (Fig. 1). This difference reflects the distinct spectral constructions of the two methods, which is the difference in reference detectability rather than in instantaneous wave amplitude.

In conclusion, the evaluation of CCW simulations can be based on three com-plementary criteria. First, consistency with observations: a model SR within the observational percentile range ($[SR_{obs}]_{5-95\%}$) indicates statistical agreement with observed variability for the same record length. Second, robustness against noise: an SR exceeding the 95th percentile of red-noise values ($[SR_{noise}]_{95\%}$) indicates that the simulated signal is unlikely to arise from stochastic variability alone. Third, bench-marking: simulations closer to the standard SR derived from long observational records can be considered more realistic. These criteria are independent rather than hierarchical. For example, a model may be consistent with observations but fall below the noise threshold, implying that the record is too short to confirm robust wave activity; or it may exceed the noise threshold but fall outside the observational range, implying robust but biased activity. Ideally, simulations should satisfy all three conditions, ensuring that they are consistent with observations, statistically robust, and com-parable to long-record observational benchmarks. This criterion is, then, applied in our case study (Sect. 3.2) to evaluate configurations and can be extended to future studies of CCW simulation fidelity.

3.2 Evaluating CCWs in a k-scale model: kilometre-scale UM

Based on the SR-based criteria from Sect. 3.1, we conduct a 40-day case study that aims to evaluate Kelvin waves and the MJO in multiple versions of the kilometre-scale UM (see Sect. 2.1.2). From our previous results, 40-day OLR data can provide marginal detection of Kelvin waves, and precipitation data can provide an adequate detection of the MJO (Fig. 1a & c). The 40-day percentile range of SR from observational data $[SR_{obs}]_{5-95\%}$, the percentile range of SR from red noise $[SR_{noise}]_{5-95\%}$, and the SR standard come from the results of Sect. 3.1.

Both OLR-based and precipitation-based SR of kilometre-scale simulations with the same configuration but different boundary conditions (different symbol types in Fig. 2) are very close to each other, which implies that the amplitude of CCWs in kilometre-scale UM has a weak sensitivity to the driving model or boundary conditions. However, this does not mean they simulate the MJO or Kelvin wave similarly since the SR reflects only the amplitude of CCW variability and ignores wave coherence, propagation characteristics, spatiotemporal organisation, east–west asymmetry, event intermittency, and phase relationships between convection, moisture, and circulation (Brune et al. 2020).

The most reasonable simulations of Kelvin waves and the MJO are found in the 5.5-km UM with convection parameterisation, and boundary model also with this parameterisation (CTC5 GAL9) (Fig. 2a & d). OLR-based SRs for Kelvin waves in all simulations fall within the observational percentile range, while only three simulations lie within the percentile range for MJO SR derived from precipitation data: CTC5 GAL9 with boundary of GLOB10 GAL9, GLOB10 GAL9, and CTC5 GAL9 with boundary of GLOB10 RAL3 (Fig. 2a & d). All other cases remain indistinguishable from red noise for the MJO, which reflects that the available record length is insufficient to demonstrate robust wave activity.

A clear link emerges between Kelvin waves and the MJO detectability. In our simulations, stronger MJO signals tend to coincide with stronger Kelvin wave activity. Previous studies in aquaplanet and GCM frameworks have similarly highlighted that MJO propagation is strongly tied to Kelvin wave strength (Heath et al. 2021). Theoretical studies suggest one possible link: MJO propagation may be facilitated by Kelvin wave–induced moisture convergence in the lower troposphere (Zhang et al. 2020). However, both phenomena may depend on the fidelity of the same underlying model physics, and any direct interaction between them would necessarily involve nonlinear processes given their spectral separation. Thus, the consistency between the three best-performing simulations across both wave types may support the dynamical connection between Kelvin waves and the MJO.

The Hovmöller diagrams further illustrate differences in wave propagation that are not captured by SR-based results. Compared with their global counterparts (Fig. 3e1&e2), kilometre-scale models generally improve the simulation of Kelvin-wave propagation speeds and amplitudes, although substantial discrepancies remain. For example, while all kilometre-scale simulations with the GAL9 boundary condition show a continuous propagation, the location of the

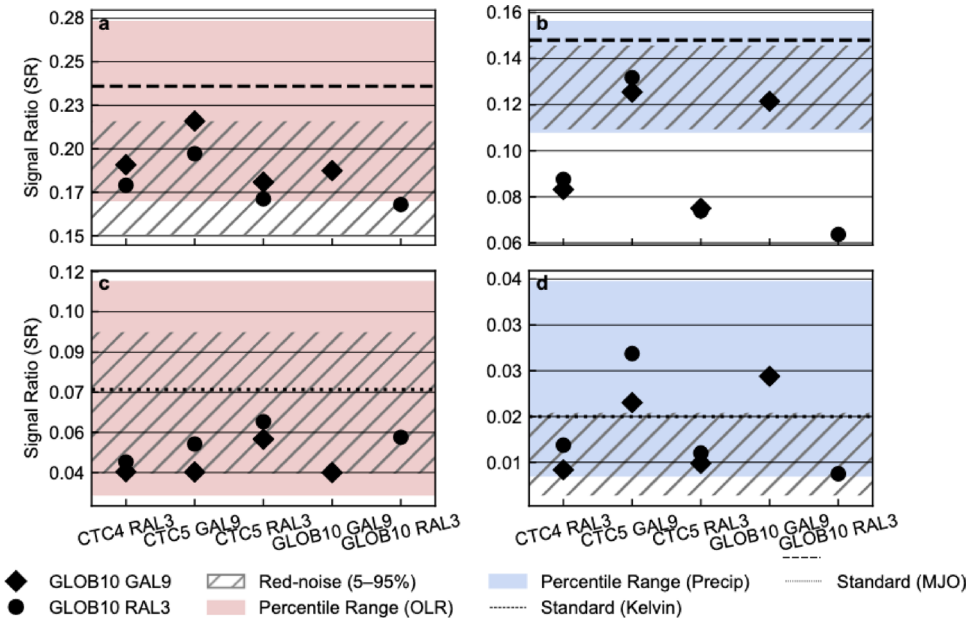


Fig. 2 SR evaluation of simulated Kelvin wave **a** & **b** and MJO **c** & **d** activity from the kilometre-scale UM. The first three sets of dots show k-UM downscaling of the tropical belt driven by the global UM having either convective parameterisation (GAL9) or explicit convection (RAL3), shown as diamond-shaped markers and dots, respectively; the three configurations differ by grid size and physics (see Sect. 2.1.2). The last two dots show results from each GLOB10 with no downscaling. OLR-based and precipitation-based SR evaluations are shown in the first **a** & **c** and the second column **b** & **d**, respectively. The first (second) row shows the SR evolution for the Kelvin wave (the MJO). The red shading shows the OLR-based, and the blue shading represents the precipitation-based, 40-day $[SR_{obs}]5-95\%$. The gray shading area with lines represents the 40-day $[SR_{noise}]5-95\%$. The long dashed line means the SR standard for Kelvin waves, and the short dashed line means the SR standard for the MJO

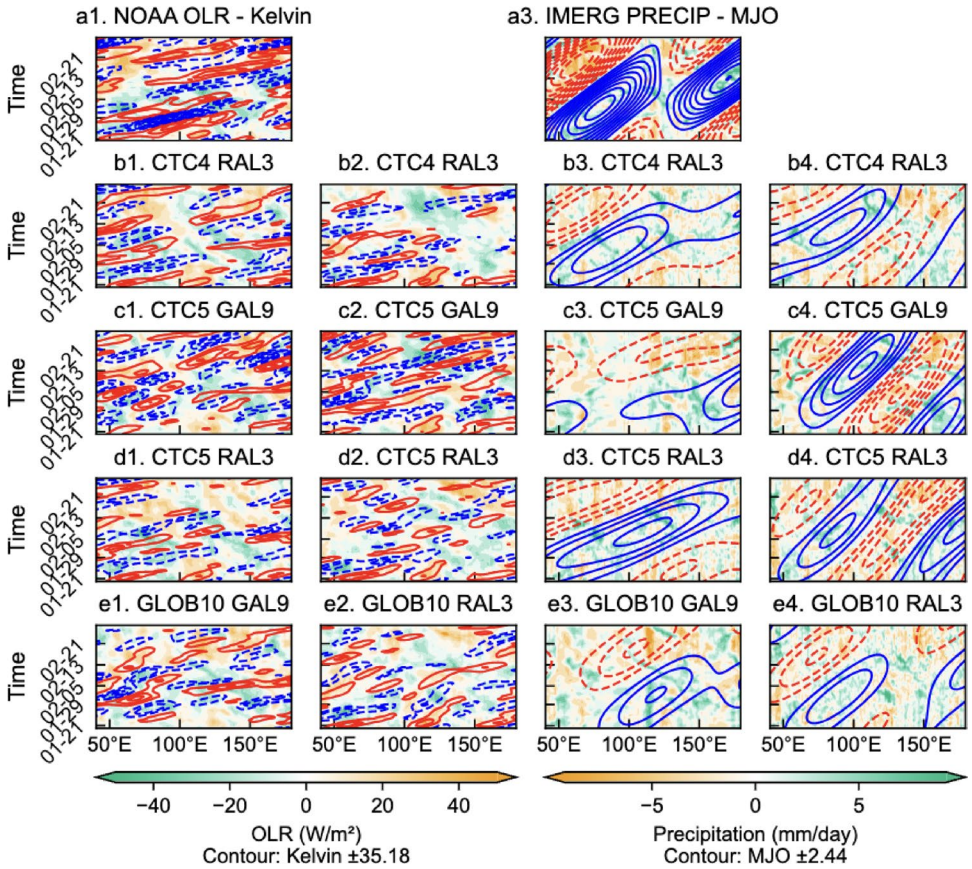


Fig. 3 Hovmöller diagram of OLR fields for the Kelvin waves (first two columns) and precipitation fields for the MJO (last two columns). The shaded background displays the Hovmöller diagrams of the original fields, and the contours represent the reconstructed field using the appropriate filter (Eq. 3). Green shading and blue contour mean positive precipitation (negative OLR), and orange shading and red contour represent negative precipitation (positive OLR). The first row shows Kelvin waves from OLR data (a1) and the MJO from precipitation (a3). The corresponding quantities from kilometre-scale UM with GLOB10 GAL9 boundary conditions are shown in (b1, c1, d1 and b3, c3, d3) with those of the GLOB10 GAL9 model itself shown in e1 and e3; similarly for kilometre-scale UM with GLOB10 RAL3 in b2, c2, d2 and b4, c4, d4, with the GLOB10 RAL3 in e2, e4. See panel titles for individual model runs

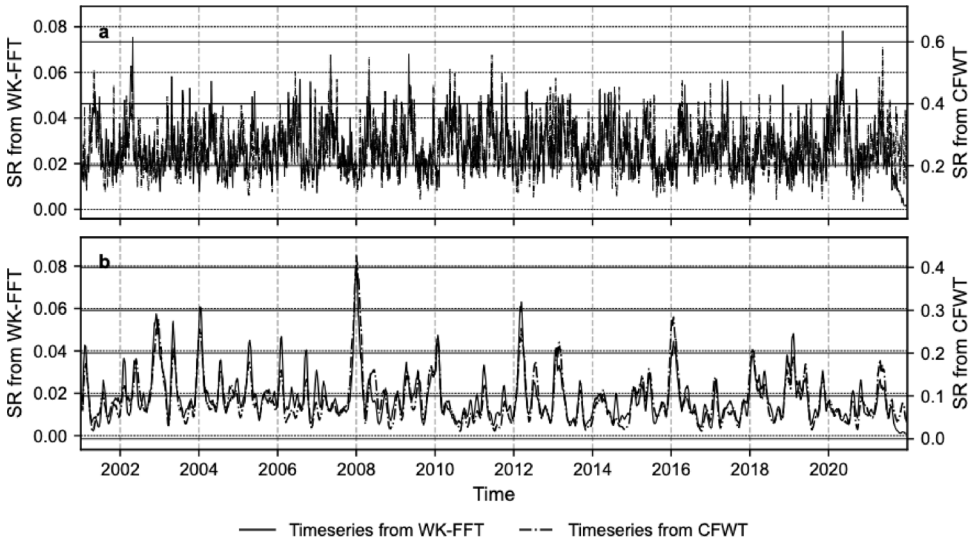


Fig. 4 SR timeseries from WK-FFT spectra (solid line) and CFWT spectra (dashed line with dot) from 2001 to 2021. SR timeseries of Kelvin waves and the MJO are shown in **a** and **b**, respectively. The left (right) y-axis shows daily SR amplitude from WK-FFT spectra (CFWT spectra). The x-axis represents days from 1st January 2001 to 31st December 2021

high-value centres exhibits variability that may arise from internal randomness as well as from resolution or configuration differences. That shows models remain sensitive in their ability to produce coherent eastward-propagating modes. In addition, MJO simulations from the UM consistently fail to reproduce the observed spatio-temporal structure, particularly as convective signals propagate across the Maritime Continent (around 100°E to 180°E). This discrepancy likely arises from the region's complex land–ocean distribution (Birch et al. 2016; Tanet al. 2020), which significantly affects the coupling between moisture, convection, and large-scale circulation (Ahn et al. 2020; Wang et al. 2022). These interactions modulate wave dynamics–such as the propagation and intensity of CCWs–as they traverse this intricate region. It is also obvious that the simulated MJO precipitation amplitude in all simulations is weaker than that observed from IMERG. While similar biases in MJO amplitude have been linked to deficiencies in moist physics in traditional GCMs Chen et al. 2022), it remains unclear whether and how such mechanisms operate in kilometre-scale models. Recent studies suggest that convection-permitting simulations at kilometre scales still exhibit biases in convective organisation (El Rafei et al. 2025), which may contribute to the reduced variability seen here.

4 Why does the CFWT perform better in detecting CCWs from short-term data?

To further explore the performance differences between WK-FFT and CFWT, the SR is calculated for every day in 20-year OLR record. The wave and total energies at time t are obtained by applying the FFT in longitude to the reconstructed wave fields, respectively. The CFWT and WK-FFT capture similar variations in Kelvin wave and MJO activity, with correlations of 0.704 and 0.893, respectively. However, the SR from CFWT is systematically higher than from WK-FFT (Fig. 4). That confirms the point from Kikuchi (2014) that the FFT-based method and the wavelet-based method, CFWT, exhibit remarkably consistent mean spectra, as both methods identify the same climatological wave modes.

The good performance of CFWT in the short-term dataset ($\leq$ one year) comes from its time–frequency localisation. The fixed-band filtering used by WK-FFT tends to smooth out transient events, reducing spectral energy when compared to the original variance (Miyakawa and Kikuchi 2018). In contrast, CFWT, by virtue of its temporal localisation, preserves variance and can resolve intermittent or few-cycle signals that FFT-based methods would otherwise average over. As shown in Fig. 4, CFWT in the daily time series produces higher wave energy amplitudes than WK-FFT, illustrating its ability to retain transient wave events. The apparently contrasting behaviour of SR between Figs. 1 and 4 does not represent a contradiction. It reflects a trade-off between spectral stability and temporal fidelity of both WK-FFT and CFWT. The higher SR reference values from WK-FFT in Fig. 1 also arise from its global Fourier averaging, which suppresses high-frequency noise and yields stable spectral ratios over fixed frequency–wavenumber bands. It makes the WK-FFT yield more conservative long-term reference estimates.

Edge treatment and variance conservation also help CFWT perform more effectively for short records or event-focused time series (like the simulations in the case study). First, CFWT explicitly addresses boundary artefacts by defining a cone of influence, unlike WK-FFT, which assumes cyclic data. Then, while variance conservation is trivial for FFT, in wavelet analysis, it requires explicit normalisation. CFWT satisfies Parseval's theorem by applying frequency- and wavenumber-dependent scaling, ensuring that spectral

energy integrates back to the time variance without bias. This guarantees that CFWT captures wave energy without altering the overall timing of events.

5 Conclusions

Motivated by a desire to quantify the simulation of convectively coupled waves (CCWs) in large-domain convection-permitting models, this study tested the ability of two published detection methods to identify CCWs in short datasets, especially those less than one year. The results suggest that a wavelet-based method, CFWT, is preferable to ordinary FFT when the dataset length is comparable to the mean wave period, and show that CFWT can work surprisingly well even when the record can barely capture a single event. We provide the shortest data length and recommended data types for robust detection of both MJO and Kelvin waves. We found that OLR better detects Kelvin waves, and precipitation better detects the MJO in records shorter than one year. CFWT can detect the Kelvin wave robustly with at least 90 days of OLR data, and the MJO with at least 30 days of precipitation data. On the other hand, 180 days of OLR data would be needed for robust detection of Kelvin waves, and 60 days of precipitation data would be required for robust detection of the MJO, even using the WK-FFT method. Although Kelvin waves have shorter intrinsic periods, their signals tend to have lower signal-to-noise ratios than the MJO within the 2–20-day frequency band, where background variability is large. As a result, longer records are required to distinguish the Kelvin signal from red noise in a statistically robust way. By contrast, the MJO's stronger, low-frequency signal emerges more clearly above background noise even in shorter segments. The time variation of detected wave activity is very similar between the two events; however, CFWT has a slight advantage because its time–frequency localization allows individual wave crests and troughs to be resolved, which is particularly useful for short datasets.

Using CFWT to screen model configurations for CCWs before running long-term simulations can save computational resources. We apply our SR-based approach to evaluate CCW in 40-day simulations from several kilometre-scale configurations of the UM atmosphere model. We find that higher resolution can improve the simulation of Kelvin waves and the MJO, although kilometre-scale simulations with explicit con-vection (RAL3) remain strongly influenced by the biases in their driving boundary conditions. Among the tested configurations, the 5.5-km UM with convection param-eterisation (CTC5 GAL9) and boundary forcing from GLOB10 GAL9 performs most consistently, reproducing both Kelvin wave and MJO signals well, which is confirmed in Hovm¨oller diagnostics.

In general, model versions producing more accurate Kelvin wave activity also pro-duce more MJO activity (both measured relative to the background variance via the SR). This reinforces the close link between Kelvin and MJO simulation skill previously reported in GCMs. However, Hovm¨oller diagnostics also highlight the persistent issues in simulating their propagation across the Maritime Continent, even in the better simulations.

It is worth noting that the CCWs identified in this study should be interpreted as spectrally consistent CCW-like signals, while the conservative nature of the detection framework implies that some physically meaningful wave variability may not be explicitly detected. On one hand, these transform-based methods rely on frequency–wavenumber characteristics rather than on explicit dynamical or structural confirmation and therefore do not uniquely determine the physical origin of the diagnosed variance. Dynamically distinct phenomena, such as Rossby waves advected eastward by background westerlies, may project into the same spectral bands as Kelvin waves, leading to spectral concurrence despite different underlying dynamics. The methods applied here thus diagnose spectrally consistent CCW-like signals rather than explicitly verifying wave structure or dynamics. While structure-based approaches could further constrain physical interpretation, they are not readily applicable to the MJO or to the limited-length, high-resolution datasets considered in this study. On the other hand, the background spectrum used to diagnose statistically significant wave signals is likely to contain a mixture of wave and non-wave variability. As a result, transform-based detection methods are conservative, in the sense that some physically meaningful wave signals may not exceed the prescribed background threshold and therefore remain undetected (Wheeler and Kiladis 1999; Roundy 2020).

In conclusion, this research quantifies the conditions under which CCWs can be robustly detected across different dataset lengths and types. While differences between methods are modest, the wavelet-based approach is shown to be particularly useful for detecting CCWs in short records, and our implementation offers a transparent and reproducible framework for evaluating model performance. The results highlight that the nature of CCWs can significantly impact detection techniques, and provide insight into how to detect CCWs robustly. This research framework and SR-based criteria can be applied to determine the optimal simulation length for learning other types of CCWs, such as Rossby waves, thereby enhancing understanding of CCWs and improving their simulations, with fixed compute resources.

Author contributions Chang Xu, Steven Sherwood, Martin Jucker, and Clemente Lopez-Bravo contributed to the study conception and

design. Material preparation, data collection, and analysis were performed by Chang Xu, Richard Jones, and Steven Sherwood. The first draft of the manuscript was written by Chang Xu, and all authors commented on previous versions of the manuscript. All authors read and approved the final manuscript.

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Data & code availability The observational datasets analysed in this study are publicly available: NOAA OLR data: https://www.ncei.noaa.gov/products/climate-data-records/ outgoing-longwave-radiation-daily; IMERG precipitation data: http://climate-cms.wikis.unsw.edu.au/FROGs. Model output from the kilometre-scale UM simulations is available from the corresponding author upon reasonable request due to storage constraints. The CFWT program developed and used in this study is openly available and archived with a DOI at https://doi.org/10.5281/zenodo.18356094. The source code is also maintained on GitHub at https://github.com/Changbaober/pyCFWT. The repository contains the core routines and example scripts for applying CFWT to detect convectively coupled waves (CCWs).

Declarations

Conflict of interest The authors have no relevant financial or non-financial interests to disclose.

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